

Assignment 6: Gesture Recognition (15P + 1P)

Goals

You know different methods for recognizing unistroke gestures and can implement them. You are familiar with LSTM neural networks. You can evaluate and compare approaches and report the results accurately. You can include robust gesture detection into simple applications.

1 Implementing the \$1 Gesture Recognizer 5P

Implement Wobbrock's \$1 gesture recognizer. You can find a description on Wobbrock's website¹. Do not use AI for code generation to implement gesture recognition, try to understand each step when implementing.

Train the recognizer with at least five gestures including **rectangle**, **circle**, **check**, **delete**, and **pigtail**. Build a little user interface with *pyglet* where users can enter gestures to test the gesture recognizer.

Score

- (2P) \$1 gesture recognizer works and is efficiently implemented
- (2P) Five gestures can be distinguished
- (1P) Gesture entry user interface

2 Comparing Gesture Recognizers 5P

Create a test dataset for unistroke gestures based on Wobbrock et al.'s unistroke gesture logs. Your dataset should contain ten gestures from each class. Train an LSTM classifier with the remaining data provided by Wobbrock et al. Then, try to systematically reduce the LSTM's parameter count. Create at least three different versions.

Then, compare accuracy and prediction times of the LSTM with different parameter counts, as well as the \$1 gesture recognizer. Report method and results in a Jupyter Notebook called *unistroke_gestures.ipynb*. Which one would you choose for a practical application, and why?

Score

- (1P) Enough data captured in the same format as \$1 gesture set
- (2P) LSTM gesture classifier works and is trained on Wobbrock et al.'s gesture logs
- (1P) Systematical comparison of different parameter counts and the \$1 recognizer
- (1P) Proper visualization and description of results

¹<https://depts.washington.edu/acelab/proj/dollar/index.html>

3 Gesture Detection Game

5P

Create a Python program called *gesture_application.py*. The program should be a very simple 2D application, such as a game, a media controller, an application quick start menu – you name it. The program can be controlled with gestures (e.g. rectangle, circle, check, delete, ...) drawn by a user.

The application should support at least three distinct gestures. Ideas for a game would be something like rock-paper-scissors, or the minigame to learn new spells from *Harry Potter* PC games², where players had to follow a specified shape. Hint: Of course you can use your code of the gesture input application of task 1 and build up on it!

Score

- (1P) Gesture input works
- (2P) Three gestures are distinguished robustly
- (2P) Functionality and aesthetics of the application

4 Bonus: Study Participation

1P

Take part in Michael's data logging study for his LSTM:

<https://calendar.app.google/2G8Jvmu88RehQykG9>

²<https://gamefabrique.com/storage/screenshots/pc/harry-potter-and-the-philosophers-stone-11.png>